ULX5M-M2 v001
Intergalaktik d.o.o.
License: CERN-OHL-S v2
Year: 2026

config
nLED_act
{GPIO}
File: config.kicad_sch

power
File: power.kicad_sch

gpdi-ram
File: ram.kicad_sch

Add RP2040 so that we do not need externa JTAG

Add inline resistors for input signals

ethernet
USB_CC1
USB_CC2
{SD}
{USB}
File: ethernet.kicad_sch

4HIGH bit SD

File: lvds.kicad_sch
lvds
{LVDS0}
{LVDS1}
PERST#
PEWAKE#
CLKREQ#
PEDET
SUSCLK
DEVSLP
LED1#

M.2 M Key
PERST#
PEWAKE#
CLKREQ#
PEDET
SUSCLK
DEVSLP
LED1#
{PCIe}
File: ../GateMate_M.2_M_Key/GateMate_M.2_M_Key/M.2 M Key.kicad_sch

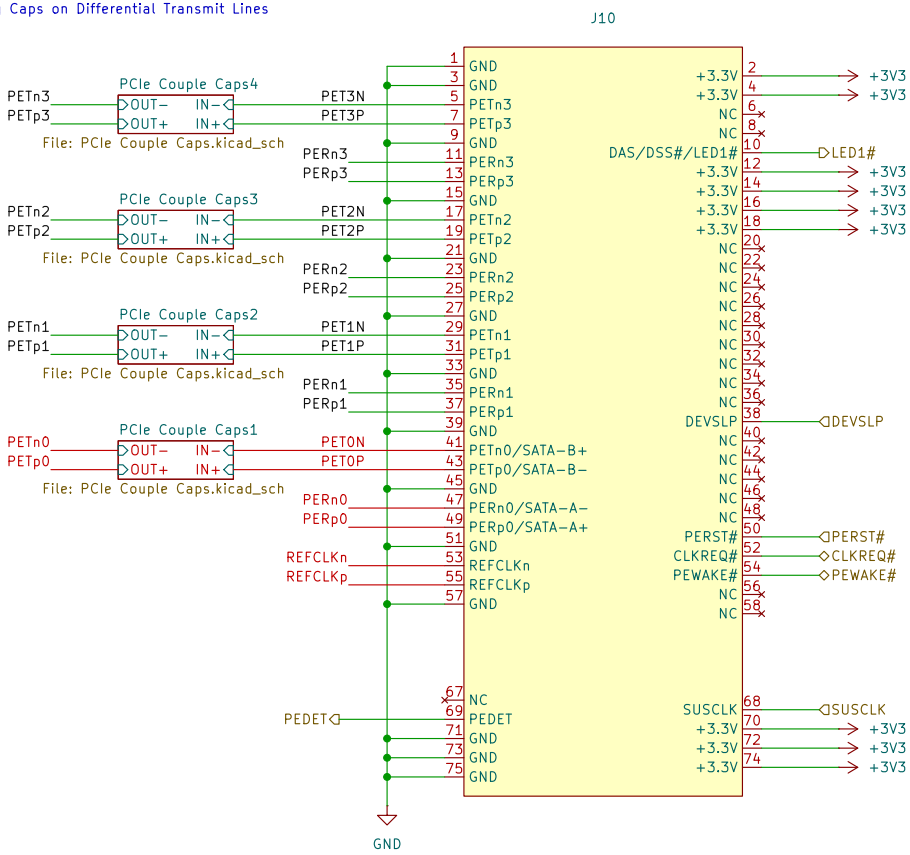
DF40C-100DP-0.4V_51_

serdes
{PCIe}
File: serdes.kicad_sch

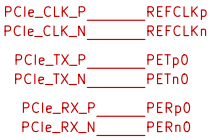
FID104 Fiducial
FID105 Fiducial
FID106 Fiducial
FID101 Fiducial
FID102 Fiducial
FID103 Fiducial

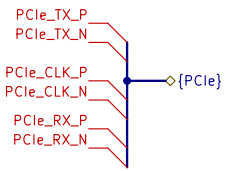
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /
File: ulx5m-m2.kicad_sch
Title: ULX5M-M2
Size: A4 | Date: 2025-06-24 | **Rev: v001**
KiCad E.D.A. 10.0.6 | Id: 1/12

Put AC Coupling Caps on Differential Transmit Lines





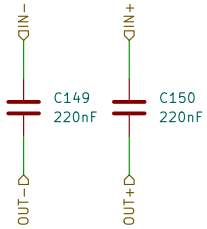




CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/
File: M.2 M Key.kicad_sch

Title: ULX5M-M2

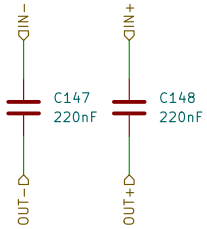
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 2/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps4/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

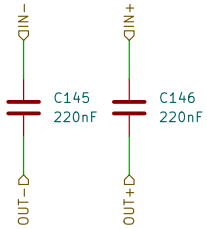
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 3/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps3/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

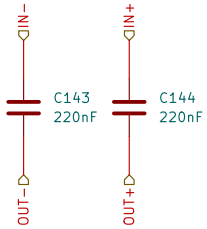
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 11/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /M.2 M Key/PCIe Couple Caps2/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 12/12



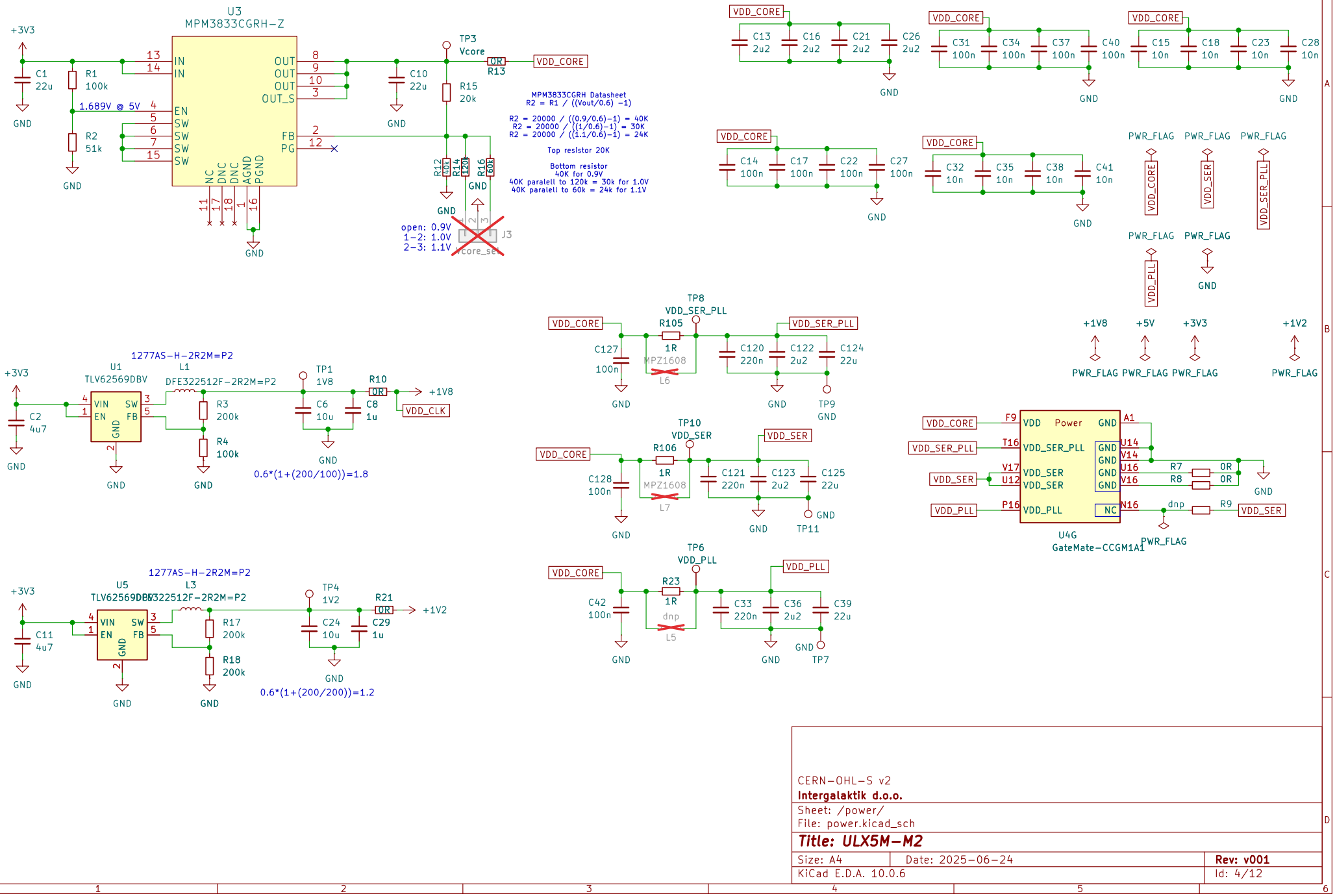
CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /M.2 M Key/PCIe Couple Caps1/
File: PCIe Couple Caps.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 13/12



CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /power/
File: power.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 4/12



Use MD0/1 PU/PD for CPHA/CPOL selection

SW1 219-2MST

U4B GateMate-CCGM1A1

U6 74LVC1G04GM,115

U7 74LVC2G00HK3-7

U8 74LVC2G00HK3-7

U9 AT25QL321-MBUE-T

J101 JTAG_EXT

J102 3V3_PSU

J4 JTAG

CERN-OHL-S v2

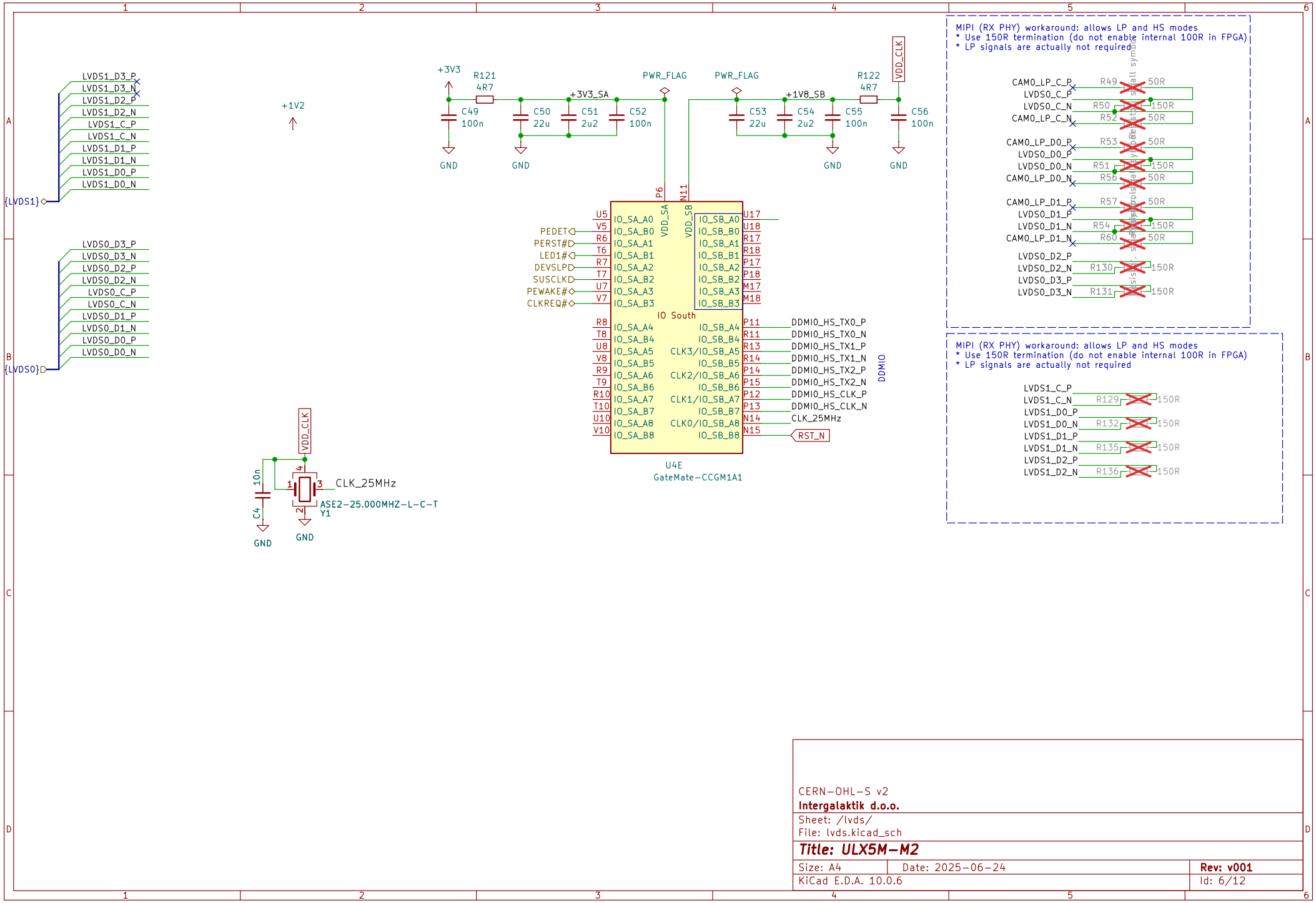
Intergalaktik d.o.o.

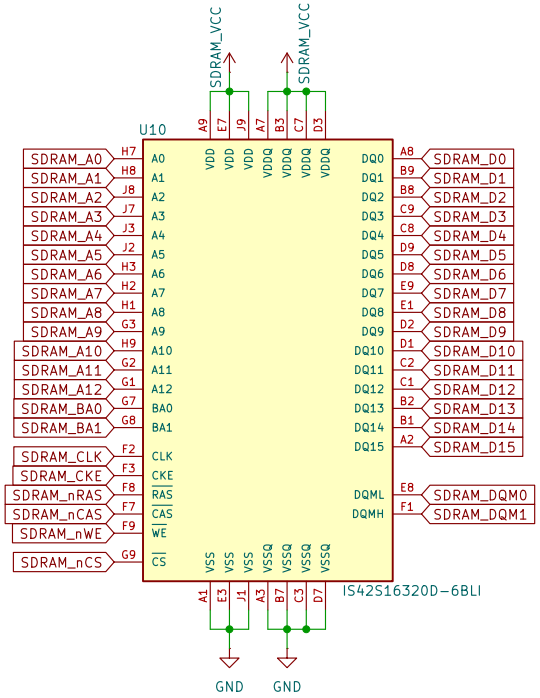
Sheet: /config/

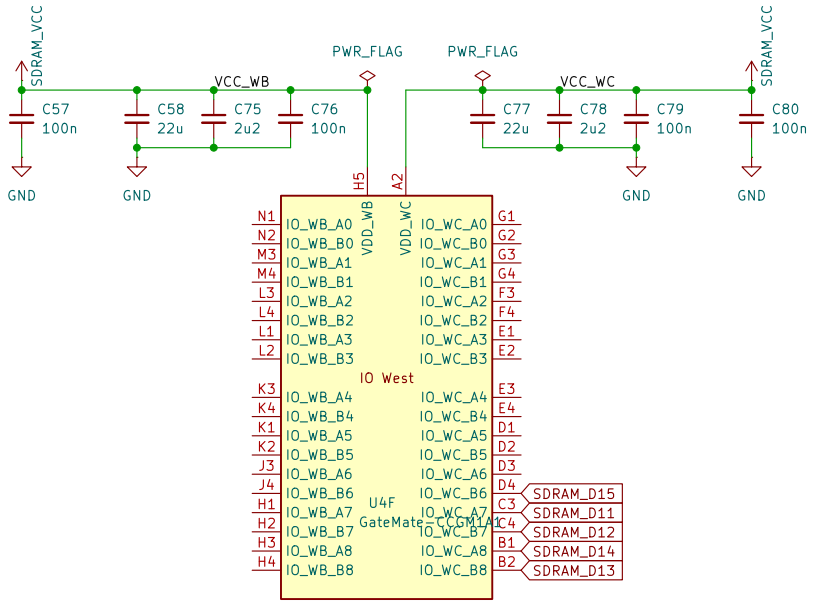
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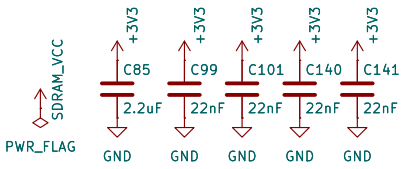
Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
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LVDS1_D3_P
LVDS1_D3_N
LVDS1_D2_P
LVDS1_D2_N
LVDS1_C_P
LVDS1_C_N
LVDS1_D1_P
LVDS1_D1_N
LVDS1_D0_P
LVDS1_D0_N
{LVDS1}
LVDS0_D3_P
LVDS0_D3_N
LVDS0_D2_P
LVDS0_D2_N
LVDS0_C_P
LVDS0_C_N
LVDS0_D1_P
LVDS0_D1_N
LVDS0_D0_P
LVDS0_D0_N
{LVDS0}
+1V2
+3V3
R121
4R7
C49
100n
C50
22u
+3V3_SA
C51
2u2
C52
100n
GND
PWR_FLAG
PWR_FLAG
+1V8_SB
R122
4R7
VDD_CLK
C53
22u
C54
2u2
C55
100n
C56
100n
P6
N11
VDD_SA
VDD_SB
U5 IO_SA_A0
V5 IO_SA_B0
R6 IO_SA_A1
T6 IO_SA_B1
R7 IO_SA_A2
T7 IO_SA_B2
U7 IO_SA_A3
V7 IO_SA_B3
PEDET
PERST#
LED1#
DEVSLP
SUSCLK
PEWAKE#
CLKREQ#
IO_SB_A0 U17
IO_SB_B0 U18
IO_SB_A1 R17
IO_SB_B1 R18
IO_SB_A2 P17
IO_SB_B2 P18
IO_SB_A3 M17
IO_SB_B3 M18
IO South
R8 IO_SA_A4
T8 IO_SA_B4
U8 IO_SA_A5
V8 IO_SA_B5
R9 IO_SA_A6
T9 IO_SA_B6
R10 IO_SA_A7
T10 IO_SA_B7
U10 IO_SA_A8
V10 IO_SA_B8
IO_SB_A4 P11 DDMIO_HS_TX0_P
IO_SB_B4 R11 DDMIO_HS_TX0_N
CLK3/IO_SB_A5 R13 DDMIO_HS_TX1_P
IO_SB_B5 R14 DDMIO_HS_TX1_N
CLK2/IO_SB_A6 P14 DDMIO_HS_TX2_P
IO_SB_B6 P15 DDMIO_HS_TX2_N
CLK1/IO_SB_A7 P12 DDMIO_HS_CLK_P
IO_SB_B7 P13 DDMIO_HS_CLK_N
IO_SB_A8 N14 CLK_25MHz
IO_SB_B8 N15 RST_N
DDMIO
U4E
GateMate-CCGM1A1
VDD_CLK
C4
10n
GND
CLK_25MHz
ASE2-25.000MHZ-L-C-T
Y1
GND
MIPI (RX PHY) workaround: allows LP and HS modes
* Use 150R termination (do not enable internal 100R in FPGA)
* LP signals are actually not required
CAM0_LP_C_P R49 50R
LVDS0_C_P
LVDS0_C_N R50 150R
CAM0_LP_C_N R52 50R
CAM0_LP_D0_P R53 50R
LVDS0_D0_P
LVDS0_D0_N R51 150R
CAM0_LP_D0_N R56 50R
CAM0_LP_D1_P R57 50R
LVDS0_D1_P
LVDS0_D1_N R54 150R
CAM0_LP_D1_N R60 50R
LVDS0_D2_P
LVDS0_D2_N R130 150R
LVDS0_D3_P
LVDS0_D3_N R131 150R
MIPI (RX PHY) workaround: allows LP and HS modes
* Use 150R termination (do not enable internal 100R in FPGA)
* LP signals are actually not required
LVDS1_C_P
LVDS1_C_N R129 150R
LVDS1_D0_P
LVDS1_D0_N R132 150R
LVDS1_D1_P
LVDS1_D1_N R135 150R
LVDS1_D2_P
LVDS1_D2_N R136 150R
CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /lvds/
File: lvds.kicad_sch
Title: ULX5M-M2
Size: A4
Date: 2025-06-24
Rev: v001
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Id: 6/12





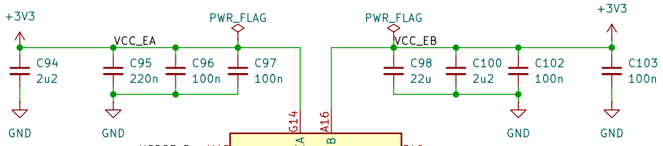


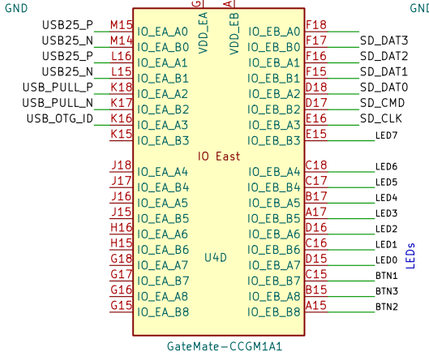
CERN-OHL-S v2
Intergalaktik d.o.o.

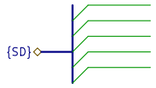
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File: ram.kicad_sch

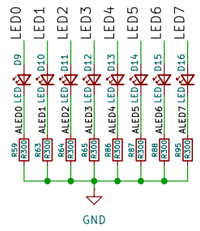
Title: ULX5M-M2

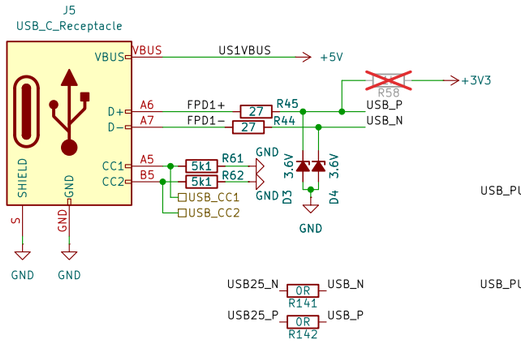
Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 7/12

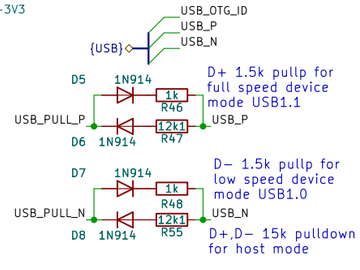


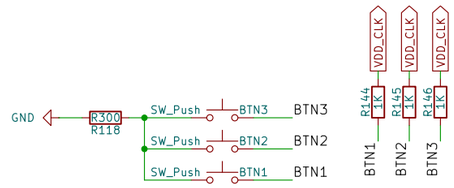








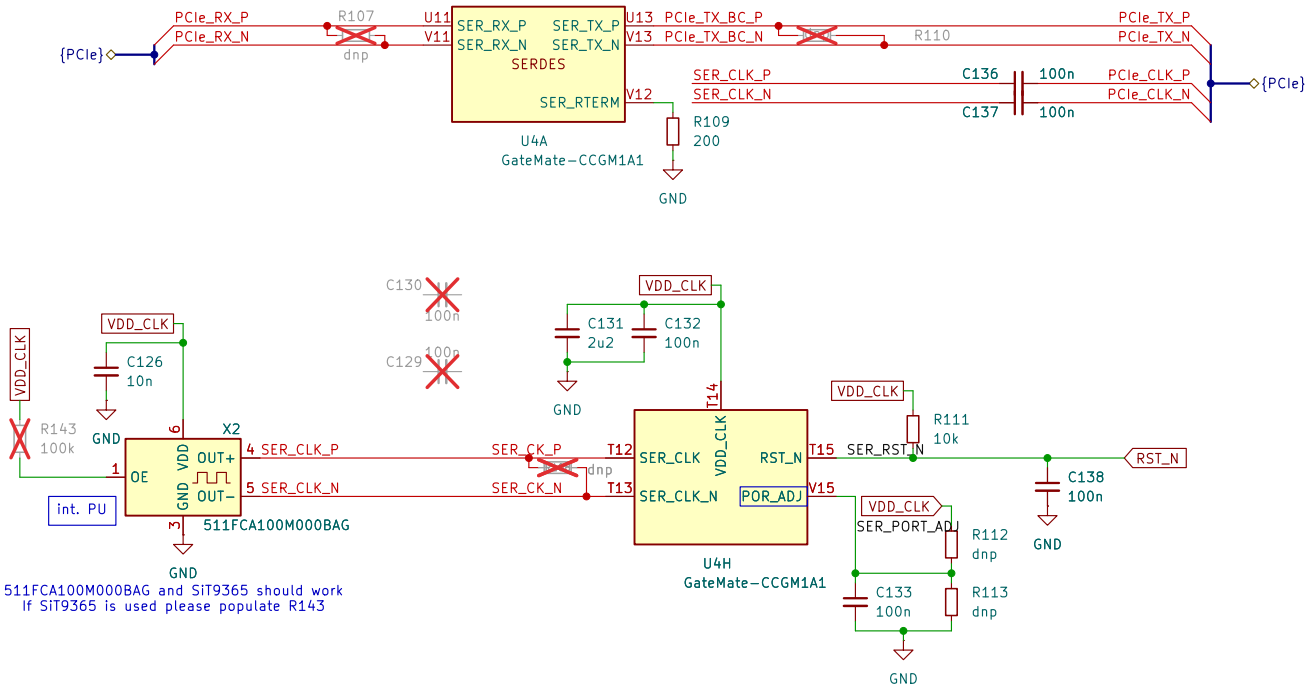




CERN-OHL-S v2
Intergalaktik d.o.o.
Sheet: /ethernet/
File: ethernet.kicad_sch

Title: ULX5M-M2

Size: A3	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 9/12



CERN-OHL-S v2

Intergalaktik d.o.o.

Sheet: /serdes/

File: serdes.kicad_sch

Title: ULX5M-M2

Size: A4	Date: 2025-06-24	**Rev: v001**
KiCad E.D.A. 10.0.6		Id: 10/12